

# A NOVEL MODIFICATION TO THE CLIMBING TEST: APPLYING IR3535, DEET AND ICARIDIN AGAINST *IXODES PERSULCATUS*

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**ABSTRACT:** Ixodid ticks are vectors of dangerous infectious diseases. Repellents are long-standing means of protecting humans against arthropod infestations. However, currently, there is no universally accepted method for evaluating the repellent effects of substances and materials on the ixodid ticks. In this study, a novel variant of the climbing test is proposed. It involves the treatment of two strips with a test substance and the subsequent evaluation of tick behaviour on human volunteers. The approach was subjected to a trial under natural conditions with taiga ticks (*Ixodes persulcatus*) collected in the Irkutskaya Oblast of Russia, near Lake Baikal. The experiments were carried out with commonly used synthetic repellents (DEET, IR3535 and icaridin) at various concentrations. Two experimental strips were used: the first 10-cm-wide strip of repellent-treated fabric was attached to the volunteer's leg, 10 cm above the foot. This strip was followed by a 10 cm buffer zone, after which the second treated strip was attached. The total climbing distance was 40 cm. Each trial was limited to 10 minutes, which corresponds to the typical time required for ticks to climb this distance in the absence of repellents. Female *I. persulcatus* were used in the experiments either on the day they were collected or the next day. A ranking system was developed to assess the repellent's efficacy. The proposed method can be used to evaluate the effectiveness of new repellent substances, as well as the protective properties of clothing treated with them.

**KEY WORDS:** *Ixodes persulcatus*, repellents, climbing test, DEET, IR3535, icaridin

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## INTRODUCTION

Among human disease, vector-borne diseases are becoming increasingly important as the habitats of many arthropods expand in response to climate change (Rocklöv and Dubrow 2020). Ticks have always held a special place among the arthropods that transmit vector-borne diseases. They are characterized by a wide variety of transmissible diseases such as Crimean-Congo hemorrhagic fever (Kuehnert *et al.* 2021), Lyme disease (Bush and Vazquez-Pertejo 2018), relapsing fever (borreliosis) (Dworkin *et al.* 2008), rickettsial diseases (e.g., spotted fever and Q fever) (Mum *et al.* 2008), tick-borne encephalitis (Czarnowska *et al.* 2024) and tularemia (Zellner and Huntley 2019). Among the main ticks carrying vector-borne diseases prevalent in North America and Eurasia, the taiga tick has been neglected by researchers despite its steadily increasing distribution (Wang *et al.* 2023). *Ixodes persulcatus* is bigger than its sister species, *I. ricinus* and *I. scapularis*, which is why repellents designed for other ticks frequently fail to affect it. Therefore, the estimation of effective concentration

required for repelling the taiga tick is an urgent task.

There are numerous strategies utilized to avoid tick attacks: the application of entomopathogenic fungi (Mastropaolo *et al.* 2017), acaricides (Eisen and Dolan 2016), repellents (Bissinger and Roe 2010), pheromones (Sonenshine 2006), tickbots (Gaff *et al.* 2015), landscape management (Stafford 2007), deer tick control applicators (Poché *et al.* 2023), bait tubes and bait boxes (Wężyk *et al.* 2024). Due to several factors—such as the effective action distance of repellents (Bissinger and Roe 2010), habitats sensitive to toxic substance treatment (van Wieren *et al.* 2016) and seasonal activity (Gray 2008)—individual self-protection using repellents/acaricides in the form of aerosol sprays (Eisen 2022; Schwartz *et al.* 2022) or impregnated clothes (Zverev *et al.* 2023) remains the most effective, ecologically friendly and cost-efficient method of defense against tick bites (Eisen 2022).

The mechanism of repellent action of various substances against ticks remains unclear (Städele

2024). That is why it is difficult to compare different repellents with each other. Therefore, the development of methods for evaluating the repellent action against ticks is an actual issue in the field of pest management. Several methods were developed to evaluate the repellent efficacy against ticks (Lingeman *et al.* 2024). Nevertheless, it is widely acknowledged that the most appropriate tests are those that most closely simulate the tick's hunting conditions. One such form of assessment is the climbing test, in which a tick is placed on a high sock, entirely impregnated with an active compound, on the leg of the human subject (Bissinger *et al.* 2011). However, since repellency is defined as movement away from the repellent source (Dethier *et al.* 1960), it is challenging to detect the repellent effect for a tick in the middle of a treated sample (Carroll *et al.* 1989). When the entire sample has been impregnated with the repellent, the continuation of climbing could be attributed to both negative geotaxis and the repellent effect. In this manuscript we propose to use two repellent barriers with a buffer zone between them. In this scenario, failing to cross at least one of the obstacles is indicative of a repellent action.

Consequently, contact-based methodologies appear to be more appropriate for the study of tick repellency. In this investigation, the objective was to conduct the study as closely to natural conditions as possible. The trials were conducted on volunteers at the forest's edge, with the majority of the experiments employing freshly collected ticks. A notable benefit of the climbing test is its capacity to emulate the natural behaviour of ticks in seeking out a suitable location on their hosts for feeding. Authors have noted that due to the difficulty in comparing the results across different testing methods, it is imperative that trials are conducted under conditions that closely resemble actual repellent use (Burtis *et al.* 2023). The present study reports the development of a modification to the climbing test for evaluating the repellent efficiency of DEET, IR3535 and icaridin against the taiga tick (*I. persulcatus*).

## MATERIALS AND METHODS

### Chemicals

DEET (*N,N*-diethyl-*meta*-toluamide) was purchased from Merck KGaA (Darmstadt, Germany). IR3535 (ethyl 3-(*N*-butylacetamido)propanoate) and icaridin (KBR 3023, butan-2-yl 2-(2-hydroxy-

ethyl)piperidine-1-carboxylate) were purchased from Upeco Ltd. (Moscow, Russia). All other reagents were of analytical grade or higher and used without further purification.

### Preparing of textile samples

Strips of white calico (10 × 50 cm, 160 g/m<sup>2</sup>, Shuiskie Sitzy LLC, Shuya, Russia) were used in the tests. The fabric was immersed in an alcohol solution until fully saturated to achieve the desired concentration of the tested substance. Preliminary laboratory tests showed that soaking time of five minutes was optimal for all compounds, regardless of their concentration. The concentrations of substances on the textile (*C*, g/m<sup>2</sup>) were calculated using the following formula:

$$C = \frac{m_s \cdot X}{100\% \cdot S},$$

where  $m_s$  is the mass (in g) of the alcohol solution absorbed by the textile (the maximum absorbency of alcohol solutions containing different concentrations of test substances was previously determined under laboratory conditions);

$X$ —substance content in the solution (w/w), %;

$S$ —textile surface area, m<sup>2</sup>.

After soaking in the solution, the textile samples were dried for 3 hours at 22–25 °C. This time frame was chosen based on preliminary findings, which have indicated that the mass of all substances becomes consistent during this period. The impregnated textiles were used on the day of their preparation.

DEET, IR3535 and icaridin were selected for the study as some of the most used repellents. The active component concentrations on the tissues were selected based on the findings of our earlier investigations (Andreev *et al.* 2024). It was done to compare the results obtained from the climbing and Petri dish assays. The following textile samples were prepared: 0.17, 1.73, 17.3, 66.0 and 110 g/m<sup>2</sup>. The standard deviation of the concentration after impregnation was <5% (n=10).

### Tick collection

Taiga ticks were collected along forest roads and forest edges near the Taltsa River, Irkutskaya Oblast, Russia (52°00'49.2" N 104°40'26.0" E), in May–June 2024, during warm and dry weather period (air temperatures of 13–23 °C and relative humidity of 50–70%), under mixed coniferous-

leaved forest canopy. Ticks were collected on white cotton cloth (flags) with no chemicals applied and transported to the laboratory in a damp gauze tape on the same day they were collected. Female ticks were identified by their body size and by the presence of red alloscuta on their backs (Estrada-Peña *et al.* 2017). The identification of microscopically-aspirated individuals was carried out after the experiment, in laboratory conditions. Ticks were used on the collection day or the next day. Wet gauze was used to store them at 4–8 °C. One to two hours before testing, ticks were placed in room temperature.

### Study of the repellent action

The repellent effect was studied on female taiga ticks. The females of *I. persulcatus* were chosen because they are significantly larger than the males, allowing for more consistent statistical results. Thirty and twenty female *I. persulcatus* were employed in the control tests and in the experiments, respectively. Tests were conducted directly in the forest, immediately after tick collection. The mobility of the ticks was checked by placing them on the hand. Every tick was used only once.

A new thigh-high sock, free from any chemical treatment, was put on the subject's leg. The testing compounds were administered in the form of two strips, each measuring 10 cm in width, which were

placed at a distance of 10 cm from the initial climbing point. The strips were separated by 10 cm (Fig. 1 B, C). In the control trials, the cotton material that had been soaked in pure ethyl alcohol and dried, was fastened to the thigh-high sock. The buffer zone (10 cm) between the two impregnated strips was intentionally organized to differentiate the assessment of the repellent action. The repellent's effectiveness was thus appraised in terms of its potential as a deterrent to ticks' climbing. Ticks lingering or looping around the leg is a sign of disorientation (this behavior was not observed in the control tests). The results for a tick were only taken into account if it passed the first 10 cm within five minutes.

The repelling effect was assessed using a rating system. The sample received a score of zero if the tick crossed both treated strips. The sample scored one point if the tick was in the second treated strip at the end of the trial, and two points if the tick was in the buffer zone. During the trial, the sample received three points if the tick failed to cross the first treated strip. Four points were given to the sample if the tick began to move but did not enter the first treated strip or if the tick spun around after coming into contact with the material and dropped below the zero mark, signifying a knockdown effect. An additional criterion for evaluating the repellent action was the probability of a tick's fall-off after contact with a treated sample.

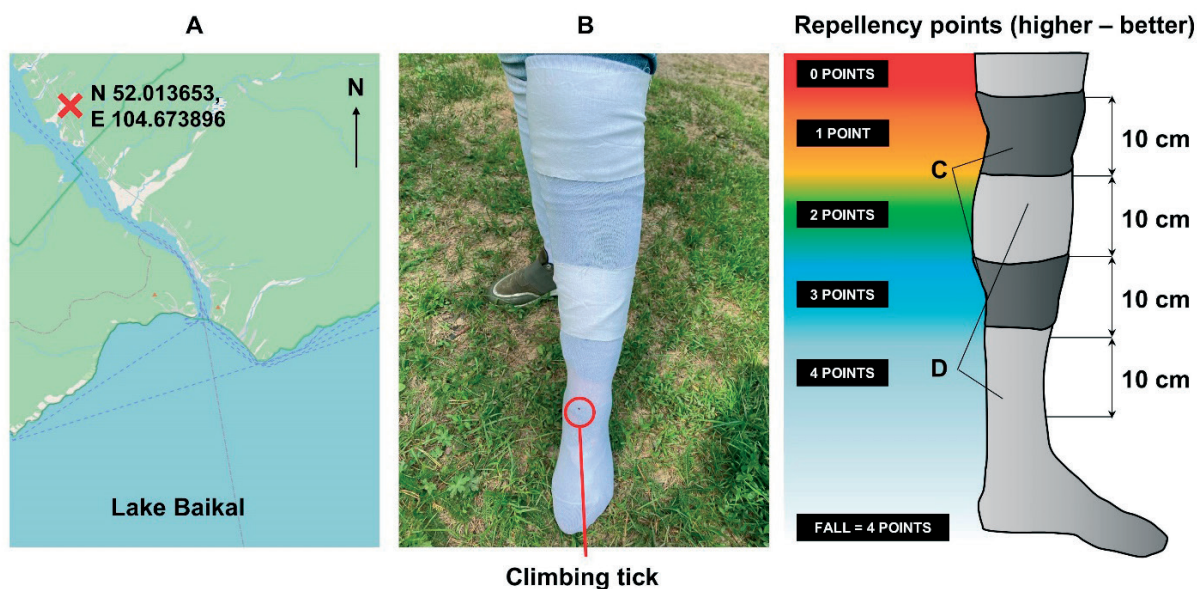


Fig. 1. Tick collection coordinates (A); photo of a thigh-high sock used in the climbing tests (B); repellent-impregnated textile strips (C); untreated textile (D).

### Data analysis

The median and the interquartile range (Q3–Q1) were calculated for each sample. The Mann–Whitney U-test was used to compare the control with the treated samples (Tsorin 2021). The null hypothesis, designated as  $H_0$ , was that the control and treated tissue tests belonged to the same general population, thereby indicating an absence of a repelling effect.

Binomial statistic was used to calculate the falling probability for all substances (Tsorin 2021). A positive result was defined as a falling tick, as this is the strongest form of repellency. The significance was set at  $P=0.05$  for all calculations. The calculations were made in Excel 2019 (Microsoft Corp.) and in OriginPro 2021 (Origin Corp.).

### RESULTS AND DISCUSSION

The use of conventional synthetic repellents facilitates the comparison of data obtained in this study with those published previously (Andreev *et al.* 2024). The following range of repellent agents was obtained, with the repellent action decreasing in the following order: icaridin, DEET, IR3535. In the present study, the insect repellents DEET and IR3535 were shown to possess a strong repellent

effect at concentrations of 17.3 g/m<sup>2</sup> and above. Table 1 shows that both chemicals received the same rates, or zero points, at concentrations of 0.173 and 1.73 g/m<sup>2</sup>. However, an analysis of the interquartile range (IQR) for IR3535 (2.25) and DEET (2) indicates that the former substance is a more potent repellent. Both IR3535 and DEET achieved a repellency score of 17.3 g/m<sup>2</sup>, but the IQR for IR3535 was zero and for DEET it was 1. This demonstrates that the proposed ranking scale emphasizes differences in a repellent's strength, even at equal concentrations of active compounds. For icaridin, weak repellent effect at 17.3 g/m<sup>2</sup> and high repellency at 66.0 g/m<sup>2</sup> were observed. In both cases, the calculated IQR was found to be elevated, indicating that the ticks that passed both treated strips, as well as the tick that did not pass the first strip, were located within the sample. Table 1 also contains the comparison of the experimental samples with the control sample using the Mann–Whitney test. The findings of this study corroborate earlier results, demonstrating that both DEET and IR3535 exhibited a repellent effect on taiga ticks at a concentration of 17.3 g/m<sup>2</sup>. Icaridin was found to demonstrate repellent properties at the concentration of 66.0 g/m<sup>2</sup>.

Table 1.  
The results of the repellent action of tissues containing different concentrations of DEET, IR3535 and icaridin on the taiga ticks using the Mann–Whitney U-test ( $n=30$  for the control test and  $n=20$  for the repellency tests).

| Substance | Concentration, g/m <sup>2</sup> | Repellency median rank | IQR (Q3–Q1) | $U_{\text{theor}}$ | $U_{\text{exp}}$ | No repellency ( $H_0$ ) / Repellent effect ( $H_1$ ) |
|-----------|---------------------------------|------------------------|-------------|--------------------|------------------|------------------------------------------------------|
| DEET      | 0.173                           | 0                      | 0           | 200                | 292              | $H_0$                                                |
|           | 1.73                            | 0                      | 2           | 200                | 245,5            | $H_0$                                                |
|           | 17.3                            | 4                      | 1           | 177                | 45               | $H_1$                                                |
|           | 66.0                            | 4                      | 0           | 177                | 43               | $H_1$                                                |
|           | 110                             | 4                      | 0           | 166                | 16,5             | $H_1$                                                |
| IR3535    | 0.173                           | 0                      | 1           | 189                | 238              | $H_0$                                                |
|           | 1.73                            | 0                      | 2.25        | 177                | 182,5            | $H_0$                                                |
|           | 17.3                            | 4                      | 0           | 200                | 31               | $H_1$                                                |
|           | 66.0                            | 4                      | 0           | 200                | 11               | $H_1$                                                |
|           | 110                             | 4                      | 0.25        | 200                | 15,5             | $H_1$                                                |
| Icaridin  | 0.173                           | 0                      | 0           | 143                | 224              | $H_0$                                                |
|           | 1.73                            | 0                      | 0           | 200                | 291              | $H_0$                                                |
|           | 17.3                            | 1                      | 4           | 166                | 146              | $H_1$                                                |
|           | 66.0                            | 4                      | 4           | 200                | 114,5            | $H_1$                                                |
|           | 110                             | 4                      | 0           | 154                | 27               | $H_1$                                                |

The data presented in Table 1 also indicate that the interquartile range (IQR) can serve as an additional parameter for assessing the repellent strength. Specifically, the presence of low median repellency values in conjunction with non-zero IQR values may be indicative of the emergence of a potent repellent effect. Consequently, the proposed approach can be utilised to effectively compare different repellents, as well as varying concentrations of active ingredients. However, this method is not without its limitations, the primary of which is its inability to generate a dose–response curve. The discrete nature of the scoring scale precludes the application of a continuous probability function.

The experimental design also allows for multiple interpretations of the repellent effect. One such interpretation involves the tick falling off after contact with the sample. The probability of a tick's falling in relation to compound concentration is presented in Fig. 2. The descent of a tick can be taken as an indication of the commencement of a knockdown effect, as it temporarily loses the capacity to adhere to the fabric. However, the observation that the ticks resumed movement after falling and in some cases

attempted to bite the researcher again 10 to 15 minutes later, suggests that this effect is reversible.

A highly significant inverse correlation was observed between the number of falls and the duration of the experiment (Spearman coefficient value  $-0.949$ ). A similar correlation was also identified in the case of DEET and IR3535, although it was found to be less pronounced (Spearman coefficient values of  $-0.500$  and  $-0.553$ , respectively). Despite the application of remarkably elevated doses of repellents in the study, none of the repellents were observed to result in a complete descent of all ticks. The estimated probability of falling for DEET was found to be 0.750, for IR3535—0.636, and for icaridin—0.500. It is noteworthy that the maximum falling probability for IR3535 was observed at a concentration of 17.3 g/m<sup>2</sup>, for DEET—at 66.0 g/m<sup>2</sup>, and for icaridin—at 110 g/m<sup>2</sup>. Therefore, IR3535 is a more effective tick repellent than DEET and icaridin. The study demonstrated that the substance exhibited a stronger repellent action at 17.3 g/m<sup>2</sup>, and that its effect was comparable to that of DEET at 66.0 g/m<sup>2</sup> (when both LCI and UCI were taken into consideration).

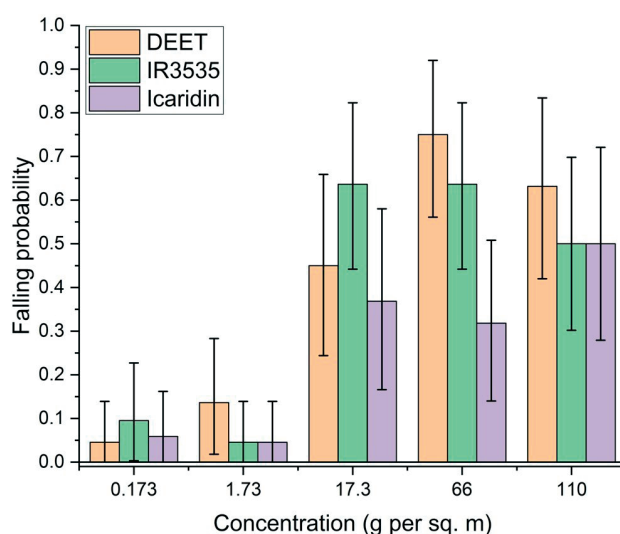


Fig. 2. The probability of ticks falling from DEET, IR3535 and icaridin

## CONCLUSION

The effectiveness of repellents is often assessed using one of three approaches: the climbing test, olfactory testing, or Petri dish experiments. The present study proposes a novel modification to the climbing test to evaluate repellency against the taiga tick. The proposed method is predicated on the execution of trials on human subjects under

natural conditions. DEET and IR3535 showed high repellency (four points) at a concentration of 17.3 g/m<sup>2</sup>, while icaridin required a much higher concentration of 66.0 g/m<sup>2</sup>. The data demonstrate that the most prevalent synthetic repellents are efficacious against the taiga tick; however, elevated concentrations are required to achieve this efficacy.

A comparison of these data with existing literature demonstrates that the effects of repellents vary across different species of ticks. Consequently, in order to devise novel personal preventive measures against ticks, it is necessary to observe the repellent action on specific tick species.

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The authors declare no conflict of interest.

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